

Course 3061

Semester III

Theory

4 Credits

Entrepreneurship and Startup Management

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 3061 as Entrepreneurship and Startup Management in Semester III for Electronics Engineering. The official SITTTR detailed source was unavailable during this cycle. With user approval, explanatory coverage is based on reliable public entrepreneurship/startup curriculum material and is not presented as recovered official SITTTR Course 3061 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional handbook placeholders. The modules transparently use public curriculum themes: entrepreneurship, idea-to-startup, management practices, support/incubators and venture proposal/exit. Reconcile precise official requirements when SITTTR content becomes accessible.

Verified source note: The official SITTTR detailed source was inaccessible in this cycle. Public sources used for the study structure are linked in References.

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Entrepreneurship and Startup Management develops the ability to identify opportunities, validate ideas, select an operating model, plan a venture and communicate a responsible business proposal. It treats entrepreneurship as disciplined problem solving rather than merely business ownership.

Malayalam support: ഈ handbook പഠിക്കുമ്പോൾ syllabus topic ആദ്യം വായിക്കുക; പിന്നീട് principle, diagram/procedure, application, common mistake എന്നിവ ക്രമമായി പഠിക്കുക.

Prerequisite foundation

No specialist prerequisite; curiosity about real-world problems, communication and basic arithmetic are useful.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain entrepreneurship, innovation and responsible opportunity recognition.
2. Develop and validate a problem–solution and customer/value proposition.
3. Apply basic management, marketing, finance and team practices to a startup.
4. Prepare a concise project proposal, pitch and responsible exit/growth plan.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO എന്നത് exam-ൽ എന്ത് ചെയ്യാൻ കഴിയണം എന്ന് പറയുന്നു. Define, explain, apply എന്നീ action words ശ്രദ്ധിക്കുക.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain entrepreneurial roles, opportunity recognition and innovation concepts.	Understand
CO2	Develop and test a basic customer-focused venture idea and business model.	Apply
CO3	Apply elementary planning, management, marketing and financial practices to a startup case.	Apply
CO4	Prepare and present a concise venture proposal with support, risk and exit considerations.	Create

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Entrepreneurship, Innovation and Opportunity	Entrepreneurial mindset, ethics, opportunity recognition, problem framing, innovation and uncertainty.	Approx. 14 hours	CO1	Module 1
2	From Idea to Startup Model	Customer discovery, value proposition, solution prototype, business model, competitors, market/price assumptions and iteration.	Approx. 16 hours	CO2	Module 2
3	Startup Management Practices	Team roles, basic operations, marketing, customer communication, cost/revenue, cash flow, compliance and risk management.	Approx. 15 hours	CO3	Module 3
4	Support Systems, Proposal and Exit	Incubators/mentors/support agencies, project proposals, pitching, funding awareness, growth options and exit/closure learning.	Approx. 15 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Entrepreneurship, Innovation and Opportunity

Entrepreneurial mindset, ethics, opportunity recognition, problem framing, innovation and uncertainty.

CO1 · Approx. 14 hours

From Idea to Startup Model

Customer discovery, value proposition, solution prototype, business model, competitors, market/price assumptions and iteration.

CO2 · Approx. 16 hours

Startup Management Practices

Team roles, basic operations, marketing, customer communication, cost/revenue, cash flow, compliance and risk management.

CO3 · Approx. 15 hours

Support Systems, Proposal and Exit

Incubators/mentors/support agencies, project proposals, pitching, funding awareness, growth options and exit/closure learning.

CO4 · Approx. 15 hours

MODULE 1

Entrepreneurship, Innovation and Opportunity

Entrepreneurial mindset, ethics, opportunity recognition, problem framing, innovation and uncertainty.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

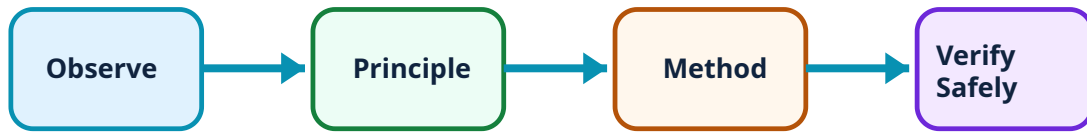
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Entrepreneurship, Innovation and Opportunity: identify the system, state its principle, follow the correct method and verify the result safely.

1. Entrepreneurship and value creation–

Entrepreneurship is the process of organising resources to create value by addressing a problem or opportunity under uncertainty. It may occur in a startup, social venture or established organisation.

Study and application method

Start with a real problem and affected user; describe the proposed value, assumptions and desired impact instead of beginning with a product label.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Start with a real problem and affected user; describe the proposed value, assumptions and desired impact instead of beginning with a product label.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: An appealing idea is not yet an opportunity until a meaningful problem and feasible value creation are evidenced.

2. Opportunity recognition and validation–

Opportunities emerge from unmet needs, changing technology, regulation, local context or process inefficiency. Validation tests assumptions with prospective users and evidence rather than personal enthusiasm.

Study and application method

Write problem hypothesis, customer segment and proposed outcome; conduct ethical interviews/observations and document what changed after evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Write problem hypothesis, customer segment and proposed outcome; conduct ethical interviews/observations and document what changed after evidence.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not treat praise from friends as market validation; seek specific behaviour, constraints and willingness evidence.

3. Responsible innovation–

Responsible ventures consider safety, inclusion, environment, privacy, legal obligations and long-term consequences alongside feasibility and revenue.

Study and application method

Map stakeholders, identify likely harms/benefits and include one mitigation or safeguard in every proposed solution.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Map stakeholders, identify likely harms/benefits and include one mitigation or safeguard in every proposed solution.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not collect or use personal customer data without clear purpose, consent and protection.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

From Idea to Startup Model

Customer discovery, value proposition, solution prototype, business model, competitors, market/price assumptions and iteration.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

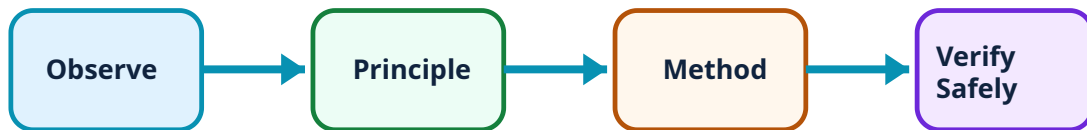
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for From Idea to Startup Model: identify the system, state its principle, follow the correct method and verify the result safely.

1. Customer and problem discovery–

Customer discovery explores users' jobs, pains and current alternatives. A useful problem statement is specific about user, context and measurable consequence.

Study and application method

Prepare open interview questions, separate observation from interpretation and synthesise evidence into a concise problem statement.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Prepare open interview questions, separate observation from interpretation and synthesise evidence into a concise problem statement.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Avoid leading questions that push people to confirm your preferred solution.

2. Value proposition and prototype–

A value proposition explains why a defined customer should prefer a proposed outcome over alternatives. A prototype may be a sketch, workflow, mock-up or limited test—not necessarily a finished product.

Study and application method

State customer, problem, promised benefit and differentiator; choose the smallest safe prototype that tests the riskiest assumption.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: State customer, problem, promised benefit and differentiator; choose the smallest safe prototype that tests the riskiest assumption.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not invest heavily in features before testing whether the problem and customer segment are real.

3. Business model and competition–

A business model links value creation, delivery and capture: customer segments, channels, key activities/resources, costs and revenue. Competition includes doing nothing and current workarounds.

Study and application method

Create a one-page model canvas, list alternatives honestly and state the assumption behind each cost/revenue estimate.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Create a one-page model canvas, list alternatives honestly and state the assumption behind each cost/revenue estimate.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not claim 'no competition' without investigating how users currently solve the problem.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Startup Management Practices

Team roles, basic operations, marketing, customer communication, cost/revenue, cash flow, compliance and risk management.

Approx. 15 hours

CO3

Understand · Apply

Learning goals

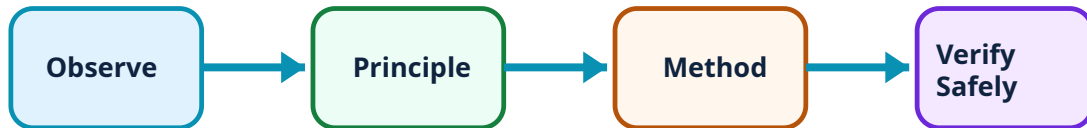
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Startup Management Practices: identify the system, state its principle, follow the correct method and verify the result safely.

1. Team and operations–

Startups coordinate people, time, suppliers and processes with limited resources. Clear roles, milestones, decision records and constructive communication reduce execution risk.

Study and application method

Define roles, deliverables, deadlines and a short review rhythm; maintain a task/risk register that the whole team can understand.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define roles, deliverables, deadlines and a short review rhythm; maintain a task/risk register that the whole team can understand.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not rely on informal verbal promises for critical ownership, payment or deliverable commitments.

2. Marketing and customer channels–

Marketing identifies a target segment, positioning, message, channel and feedback loop. Early-stage marketing prioritises learning about customer response over broad unmeasured promotion.

Study and application method

Define target segment and channel, write a measurable call to action, track response and revise based on evidence.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define target segment and channel, write a measurable call to action, track response and revise based on evidence.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not confuse social-media reach with conversion, retention or customer value.

3. Finance, compliance and risk–

Basic startup finance tracks cost, price, revenue, cash timing and runway. Compliance, intellectual property, contracts and tax/legal obligations need appropriate authoritative guidance.

Study and application method

Build a transparent assumptions sheet for costs/revenue/cash timing, identify top risks and seek qualified professional advice for legal or regulatory decisions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Build a transparent assumptions sheet for costs/revenue/cash timing, identify top risks and seek qualified professional advice for legal or regulatory decisions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Never present financial projections as certainty; label assumptions and uncertainty clearly.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Support Systems, Proposal and Exit

Incubators/mentors/support agencies, project proposals, pitching, funding awareness, growth options and exit/closure learning.

Approx. 15 hours

CO4

Understand · Apply

Learning goals

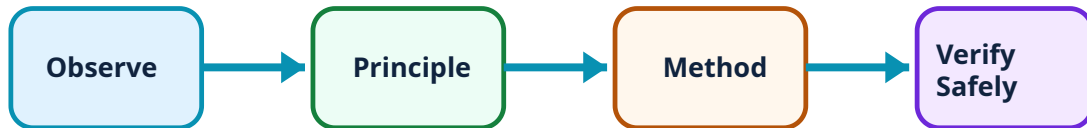
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Support Systems, Proposal and Exit: identify the system, state its principle, follow the correct method and verify the result safely.

1. Support agencies and incubation–

Incubators, mentors, maker spaces, institutions and appropriate public/private programmes can offer workspace, feedback, training, connections or finance readiness. Support should match the venture stage and needs.

Study and application method

List the venture's current need, eligibility, expected support and required commitment before approaching an agency or mentor.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the venture's current need, eligibility, expected support and required commitment before approaching an agency or mentor.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Verify programme terms, data rights and financial obligations before sharing proprietary information.

2. Project proposal and pitch–

A venture proposal explains the problem, customer, solution, evidence, model, team, milestones, resources and risks. A pitch communicates the essential story clearly to a defined audience.

Study and application method

Use problem → evidence → solution → model → plan → ask structure; rehearse within time, cite evidence and anticipate questions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Use problem → evidence → solution → model → plan → ask structure; rehearse within time, cite evidence and anticipate questions.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not use exaggerated market claims or hide risk; credibility depends on transparency.

3. Growth, pivot and exit–

A venture may grow, pivot, partner, transfer, close or exit through planned alternatives. Learning from evidence and documenting decisions makes a change in direction responsible rather than a failure.

Study and application method

Define trigger metrics, decision owner and next options; retain records and meet obligations to customers/team when changing or closing operations.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define trigger metrics, decision owner and next options; retain records and meet obligations to customers/team when changing or closing operations.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ആദ്യം concept എന്താണ് എന്ന് മനസ്സിലാക്കുക. ശേഷം diagram / circuit / formula / procedure ശ്രദ്ധിച്ച് പഠിക്കുക. അവസാനമായി result അല്ലെങ്കിൽ application സ്വന്തം വാക്കുകളിൽ പറയാൻ പരിശീലിക്കുക. Technical terms English-ൽ തന്നെ ഉപയോഗിക്കുക.

Common mistake / precaution: Do not scale operations before validating unit economics, delivery capability and customer retention.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Entrepreneurship, Innovation and Opportunity

Use the concept flow to revise observation, principle, method and verification.

Module 2: From Idea to Startup Model

Use the concept flow to revise observation, principle, method and verification.

Module 3: Startup Management Practices

Use the concept flow to revise observation, principle, method and verification.

Module 4: Support Systems, Proposal and Exit

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Interview prospective users about a locally relevant problem and document evidence without proposing a solution first.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Create a one-page value proposition and business model canvas with clearly labelled assumptions.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Prepare a simple cost/revenue/cash-timing sheet and a risk register for a student venture.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Write and deliver a three-minute evidence-based venture pitch with an explicit next-step request.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure		
Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Entrepreneurship, Innovation and Opportunity

Explain Entrepreneurship and value creation and state one correct method, application or precaution.–

Entrepreneurship is the process of organising resources to create value by addressing a problem or opportunity under uncertainty. It may occur in a startup, social venture or established organisation.

Answer framework: Start with a real problem and affected user; describe the proposed value, assumptions and desired impact instead of beginning with a product label.

Important point: An appealing idea is not yet an opportunity until a meaningful problem and feasible value creation are evidenced.

Explain Opportunity recognition and validation and state one correct method, application or precaution.–

Opportunities emerge from unmet needs, changing technology, regulation, local context or process inefficiency. Validation tests assumptions with prospective users and evidence rather than personal enthusiasm.

Answer framework: Write problem hypothesis, customer segment and proposed outcome; conduct ethical interviews/observations and document what changed after evidence.

Important point: Do not treat praise from friends as market validation; seek specific behaviour, constraints and willingness evidence.

Explain Responsible innovation and state one correct method, application or precaution.–

Responsible ventures consider safety, inclusion, environment, privacy, legal obligations and long-term consequences alongside feasibility and revenue.

Answer framework: Map stakeholders, identify likely harms/benefits and include one mitigation or safeguard in every proposed solution.

Important point: Do not collect or use personal customer data without clear purpose, consent and protection.

Module 2 — From Idea to Startup Model

Explain Customer and problem discovery and state one correct method, application or precaution.—

Customer discovery explores users' jobs, pains and current alternatives. A useful problem statement is specific about user, context and measurable consequence.

Answer framework: Prepare open interview questions, separate observation from interpretation and synthesise evidence into a concise problem statement.

Important point: Avoid leading questions that push people to confirm your preferred solution.

Explain Value proposition and prototype and state one correct method, application or precaution.—

A value proposition explains why a defined customer should prefer a proposed outcome over alternatives. A prototype may be a sketch, workflow, mock-up or limited test—not necessarily a finished product.

Answer framework: State customer, problem, promised benefit and differentiator; choose the smallest safe prototype that tests the riskiest assumption.

Important point: Do not invest heavily in features before testing whether the problem and customer segment are real.

Explain Business model and competition and state one correct method, application or precaution.—

A business model links value creation, delivery and capture: customer segments, channels, key activities/resources, costs and revenue. Competition includes doing nothing and current workarounds.

Answer framework: Create a one-page model canvas, list alternatives honestly and state the assumption behind each cost/revenue estimate.

Important point: Do not claim 'no competition' without investigating how users currently solve the problem.

Module 3 — Startup Management Practices

Explain Team and operations and state one correct method, application or precaution.—

—

Startups coordinate people, time, suppliers and processes with limited resources. Clear roles, milestones, decision records and constructive communication reduce execution risk.

Answer framework: Define roles, deliverables, deadlines and a short review rhythm; maintain a task/risk register that the whole team can understand.

Important point: Do not rely on informal verbal promises for critical ownership, payment or deliverable commitments.

Explain Marketing and customer channels and state one correct method, application or precaution.–

Marketing identifies a target segment, positioning, message, channel and feedback loop. Early-stage marketing prioritises learning about customer response over broad unmeasured promotion.

Answer framework: Define target segment and channel, write a measurable call to action, track response and revise based on evidence.

Important point: Do not confuse social-media reach with conversion, retention or customer value.

Explain Finance, compliance and risk and state one correct method, application or precaution.–

Basic startup finance tracks cost, price, revenue, cash timing and runway. Compliance, intellectual property, contracts and tax/legal obligations need appropriate authoritative guidance.

Answer framework: Build a transparent assumptions sheet for costs/revenue/cash timing, identify top risks and seek qualified professional advice for legal or regulatory decisions.

Important point: Never present financial projections as certainty; label assumptions and uncertainty clearly.

Module 4 — Support Systems, Proposal and Exit

Explain Support agencies and incubation and state one correct method, application or precaution.–

Incubators, mentors, maker spaces, institutions and appropriate public/private programmes can offer workspace, feedback, training, connections or finance readiness. Support should match the venture stage and needs.

Answer framework: List the venture's current need, eligibility, expected support and required commitment before approaching an agency or mentor.

Important point: Verify programme terms, data rights and financial obligations before sharing proprietary information.

Explain Project proposal and pitch and state one correct method, application or precaution.–

A venture proposal explains the problem, customer, solution, evidence, model, team, milestones, resources and risks. A pitch communicates the essential story clearly to a defined audience.

Answer framework: Use problem → evidence → solution → model → plan → ask structure; rehearse within time, cite evidence and anticipate questions.

Important point: Do not use exaggerated market claims or hide risk; credibility depends on transparency.

Explain Growth, pivot and exit and state one correct method, application or precaution.–

A venture may grow, pivot, partner, transfer, close or exit through planned alternatives. Learning from evidence and documenting decisions makes a change in direction responsible rather than a failure.

Answer framework: Define trigger metrics, decision owner and next options; retain records and meet obligations to customers/team when changing or closing operations.

Important point: Do not scale operations before validating unit economics, delivery capability and customer retention.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Entrepreneurship, Innovation and Opportunity.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.

3. State one key definition from Module 2: From Idea to Startup Model.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Startup Management Practices.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Support Systems, Proposal and Exit.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Entrepreneurial mindset, ethics, opportunity recognition, problem framing, innovation and uncertainty..
2. Develop a complete answer or practical plan for: Customer discovery, value proposition, solution prototype, business model, competitors, market/price assumptions and iteration..
3. Develop a complete answer or practical plan for: Team roles, basic operations, marketing, customer communication, cost/revenue, cash flow, compliance and risk management..
4. Develop a complete answer or practical plan for: Incubators/mentors/support agencies, project proposals, pitching, funding awareness, growth options and exit/closure learning..

LAST-MILE REVISION

Quick Revision

Module 1: Entrepreneurship, Innovation and Opportunity

Entrepreneurial mindset, ethics, opportunity recognition, problem framing, innovation and uncertainty.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: From Idea to Startup Model

Customer discovery, value proposition, solution prototype, business model, competitors, market/price assumptions and iteration.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Startup Management Practices

Team roles, basic operations, marketing, customer communication, cost/revenue, cash flow, compliance and risk management.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Support Systems, Proposal and Exit

Incubators/mentors/support agencies, project proposals, pitching, funding awareness, growth options and exit/closure learning.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Entrepreneurship and value creation	Entrepreneurship is the process of organising resources to create value by addressing a problem or opportunity under uncertainty.
Opportunity recognition and validation	Opportunities emerge from unmet needs, changing technology, regulation, local context or process inefficiency.
Responsible innovation	Responsible ventures consider safety, inclusion, environment, privacy, legal obligations and long-term consequences alongside feasibility and revenue.
Customer and problem discovery	Customer discovery explores users' jobs, pains and current alternatives.
Value proposition and prototype	A value proposition explains why a defined customer should prefer a proposed outcome over alternatives.
Business model and competition	A business model links value creation, delivery and capture: customer segments, channels, key activities/resources, costs and revenue.
Team and operations	Startups coordinate people, time, suppliers and processes with limited resources.
Marketing and customer channels	Marketing identifies a target segment, positioning, message, channel and feedback loop.
Finance, compliance and risk	Basic startup finance tracks cost, price, revenue, cash timing and runway.
Support agencies and incubation	Incubators, mentors, maker spaces, institutions and appropriate public/private programmes can offer workspace, feedback, training, connections or finance readiness.
Project proposal and pitch	A venture proposal explains the problem, customer, solution, evidence, model, team, milestones, resources and risks.
Growth, pivot and exit	A venture may grow, pivot, partner, transfer, close or exit through planned alternatives.

LEARNING RESOURCES

References

Recommended entrepreneurship references

1. Peter Drucker, Innovation and Entrepreneurship.
2. Eric Ries, The Lean Startup.
3. Alexander Osterwalder and Yves Pigneur, Business Model Generation.

Approved public entrepreneurship sources

- <https://vpmpce.wordpress.com/entrepreneurship-start-ups-4300021/>
- <https://www.washington.edu/students/crscat/entre.html>

The listed public sources support this study handbook while official Course 3061 content is unavailable; they do not replace an official detailed syllabus.